

VoiceGuard: An Evaluation-Honest, Multi-Signal Decision Layer for Real-Time Voice-Clone Detection in High-Stakes Telephone Calls

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Abstract—Published speech anti-spoofing countermeasures report sub-1% Equal Error Rate (EER) on the ASVspoof 2019 benchmark, yet prior work shows this performance collapses on audio the models were not trained on. We make three contributions toward closing the gap between benchmark and deployment for telephone-based voice-clone fraud. First, we identify and close a silence-duration/loudness shortcut in the standard ASVspoof 2019 LA evaluation pipeline: a classifier trained on unprotected features scored 13-15% EER using only the non-speech regions of dev clips (chance is 50%), meaning part of its apparent accuracy came from an artifact, not synthesis. We propose a reusable anti-shortcut front-end and a permanent silence-ablation gate, and re-evaluate a published countermeasure (AASIST) honestly: its cross-dataset (In-the-Wild) EER is 35.8%, far from its published in-domain figure. Second, motivated by the finding that no single acoustic detector we tested generalizes reliably, we design and build VoiceGuard, a real-time decision layer that fuses acoustic spoof detection with speaker-consistency (WavLM x-vector), prosodic/behavioral biometrics, and call-context signals into a policy-driven ALLOW/VERIFY/ESCALATE decision, with a feature-only, privacy-preserving audit trail. We show a concrete case where the acoustic signal alone (29/100 risk) would pass a cloned call, while the fused system escalates it (68/100) because speaker consistency fails. Third, we report a pilot cross-lingual study across five Indian languages showing synthetic-speech recall ranging from 7.5% (English) to 85% (Marathi) for the same English-trained detector -- evidence that multilingual robustness must be measured, not assumed. We report every result with full provenance and state our limitations explicitly.

Index Terms—audio anti-spoofing, voice-clone detection, speaker verification, sensor fusion, generalization, evaluation methodology, telephony fraud, multilingual speech, ASVspoof

I. INTRODUCTION

Low-cost, real-time voice cloning has moved from a research curiosity to a fraud vector. A few seconds of a target's speech, obtained from a social-media clip or a previous call, is enough for current text-to-speech (TTS) and voice-conversion (VC) systems to produce speech that is convincing over a telephone channel. Financial institutions and call centers that rely on voice as a soft authentication factor -- for transaction confirmation, beneficiary changes, or agent verification -- are directly exposed. This is the problem framing behind Smart India Hackathon problem statement SIH26104, which asks for a real-time system that detects cloned or synthetic speech and verifies the claimed speaker and measures behavioral anomalies and recommends an action, before a transaction, under a privacy-preserving design.

The academic literature on synthetic-speech detection reports numbers that suggest the detection half of this problem is close to solved: countermeasures trained and evaluated on the ASVspoof 2019 Logical Access (LA) corpus routinely report Equal Error Rates (EER) below 1% [2]. Independent replications tell a different story. Müller et al. show that models trained on ASVspoof-family data lose up to an order of magnitude in accuracy on found ("In-the-Wild") audio [1], and a follow-up study attributes most of this gap to distributional difference between generators rather than to any generator being intrinsically "harder" to detect [5]. Our own experiments, reported in this paper, reproduce and extend this finding, and additionally show that part of the historically reported in-domain accuracy on ASVspoof 2019 LA itself is an artifact of the corpus, not of synthesis.

This paper reports what we built in response to that finding: not a better single classifier, but (i) an evaluation protocol that

removes a known shortcut and makes every subsequent number trustworthy, and (ii) a decision system, VoiceGuard, that treats acoustic spoof detection as one of several weak, partly independent signals rather than as the whole system, fuses them into an auditable, policy-driven recommendation, and keeps a human in the loop. We view this as the practically correct response to a field where no detector we evaluated -- including a second, independently published deepfake-audio classifier we tested for comparison -- reliably caught a modern neural-TTS clone in isolation.

Contributions.

- We identify a silence-duration/loudness shortcut in the ASVspoof 2019 LA evaluation pipeline (Section III-A) and propose an anti-shortcut front-end plus a permanent silence-ablation gate as a reusable protocol, stamped with a reproducible fingerprint on every reported number.
- We re-evaluate a published countermeasure (AASIST) through this protocol and report honest in-domain, unseen-attack, per-attack, and cross-dataset numbers (Section V), including a silence-only ablation confirming the shortcut is closed.
- We design and implement VoiceGuard (Section III), a four-signal fusion architecture (synthetic-voice, speaker consistency, prosody/behavior, call context) with a scenario-aware policy engine, a feature-only audit log, and a REST/WebSocket API with an operator console, and demonstrate a worked case where fusion recovers a miss by the acoustic signal alone (Section V-E).
- We report a pilot multilingual study, IndicCall-Eval, across five Indian languages, finding an English-trained detector's synthetic-speech recall varies from 7.5% to 85% by target language (Section V-F).

- We report, as a distinct contribution to reproducibility, the concrete engineering failures and fixes encountered while building this system (Section VII), several of which are not specific to our codebase.

The remainder of this paper is organized as follows. Section II surveys related work. Section III describes the VoiceGuard architecture, including the anti-shortcut front-end and the threat model. Section IV describes datasets and metrics. Section V reports results. Section VI discusses findings and limitations. Section VII reports lessons learned. Section VIII outlines future work, and Section IX concludes.

II. RELATED WORK

A. Spoofing Countermeasures and Challenges

The ASVspoof challenge series established the standard corpora and protocols for logical-access (TTS/VC), physical-access (replay), and deepfake countermeasure research. ASVspoof 2019 LA [3] remains the most widely used training/evaluation corpus for logical-access countermeasures; ASVspoof 5 [4] scales this to crowdsourced data from roughly 2,000 speakers and 32 attack algorithms, and adds adversarial attacks. The Audio Deep synthesis Detection (ADD) challenge series targets low-quality, partially fake, and adversarially generated audio under more realistic noise conditions [12]. AASIST [2], the backbone acoustic detector we use, integrates spectral and temporal graph attention over a raw-waveform front-end and is, at the time of writing, a standard strong baseline; RawBoost [6] is a widely adopted raw-waveform augmentation method for this family of models.

B. Generalization Failures

A growing body of work shows that in-domain accuracy on ASVspoof-style benchmarks does not predict real-world performance. Müller et al. [1] evaluate countermeasures trained on ASVspoof 2019 against an independently collected “In-the-Wild” set of celebrity and political speech and report large relative degradations. A follow-up study decomposes this gap and finds it is driven mainly by distributional difference between training and test generators, not by harder acoustics [5]. Pham et al. [11] survey the wider deepfake-speech-detection literature and reach a similar conclusion about generalization as the field’s central open problem. Our work reproduces this generalization gap on a model we did not train (Section V) and additionally shows that the in-domain half of the comparison can itself be inflated by a corpus artifact -- a failure mode that, to our knowledge, is not addressed by the works above, all of which evaluate on the corpus as published.

C. Speaker Verification and Voice Biometrics

Modern speaker verification uses learned embeddings -- x-vectors [8] and, more recently, embeddings drawn from large self-supervised speech models such as WavLM [7] -- compared by cosine similarity against an enrollment utterance.

Commercial voice biometric and telephony-liveness products apply this idea operationally, but published, peer-reviewed detail on their fusion logic and evaluation methodology is limited. VoiceGuard uses a WavLM x-vector as one input to an explicit, inspectable fusion and policy layer rather than as a standalone accept/reject gate.

D. Multilingual and Low-Resource Synthetic Speech

Nearly all widely used spoofing corpora and countermeasures are English-centric. IndicSynth [10] is a large, recent multilingual synthetic-speech resource spanning 12 low-resource Indian languages, explicitly motivated by the observation that detectors trained on one language transfer poorly, and in an uneven, unpredictable way, to others. Our IndicCall-Eval pilot (Section V-F) is far smaller in scale than IndicSynth but is, to our knowledge, one of few reported measurements of an English-trained countermeasure’s synthetic-recall asymmetry across Indian languages specifically in a call-center/telephony framing, and we position it as a motivating pilot for exactly the kind of larger, consented resource IndicSynth provides.

E. Positioning

The works above chiefly optimize or evaluate a single detector against a fixed benchmark. Our contribution is orthogonal: we treat the benchmark itself as a thing to be audited (Section III-A), and we treat detection as one signal inside an auditable security-decision system (Section III) rather than as a standalone accept/reject classifier. This framing follows from an empirical premise -- that no single detector we evaluated generalizes reliably enough to be trusted alone -- rather than from a claim of a new state-of-the-art detector.

III. THE VOICEGUARD SYSTEM

We consider a live or recently recorded telephone call in which a caller requests an action of consequence -- confirming a payment, changing a beneficiary, or resetting a credential. The defender observes an audio stream from the caller, an enrollment reference for the claimed identity (if any), and call metadata (caller-ID match, requested amount, channel, prior flags). The attacker’s goal is to have the system accept an instruction while impersonating the account holder, via a synthetic or replayed voice. We do not assume a single acoustic classifier can solve this reliably -- Section V shows that even a strong published detector, evaluated honestly, misses a large fraction of unseen attacks. We instead pose the problem as a bounded-risk decision: given signals for synthetic voice, speaker consistency, prosody, and context, produce an action in {ALLOW, VERIFY, ESCALATE} with a human-readable justification, under scenario-dependent risk tolerance and a latency budget compatible with a live call. The design goal is a bounded, explainable, auditable decision that degrades gracefully when any one signal is weak, absent, or wrong -- not zero missed detections, which no component here achieves.

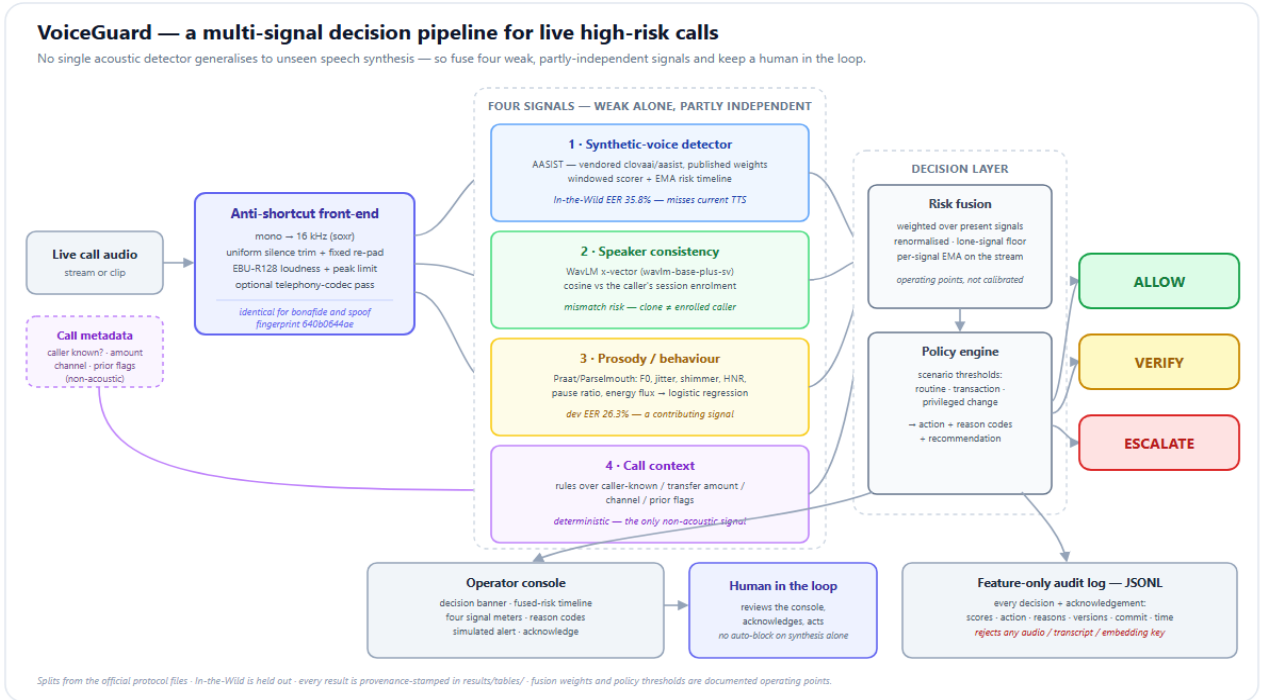


Fig. 1. VoiceGuard architecture. Call audio passes through one anti-shortcut front-end (Section III-A), shared by every downstream path. Four signals are computed -- three from audio, one from call metadata -- and fused with a lone-signal floor (Section III-F). A scenario-aware policy maps the fused score to ALLOW/VERIFY/ESCALATE with reason codes, logged to a feature-only audit trail and surfaced in an operator console.

A. Anti-Shortcut Evaluation Front-End

Early in this project, a classifier trained on handcrafted acoustic features and evaluated on the ASVspoof 2019 LA dev split scored an apparently strong 1.83% EER. As a sanity check, we re-scored the same model using only the leading and trailing non-speech regions of each dev clip -- audio that should carry no information about whether the speaker is genuine or synthetic. The model scored 13-15% EER on this silence-only input, far below the 50% expected of an uninformative signal. Bona fide and spoof clips in ASVspoof 2019 LA differ systematically in leading/trailing silence duration and loudness, almost certainly as a side effect of how the corpus was assembled and post-processed by different systems; a model can partially solve the labeling task from this artifact alone, inflating its apparent accuracy on synthesis detection.

We close this shortcut with a mandatory front-end applied identically to every clip in every split, for both classes, at feature-extraction, training, and inference time: (1) resample to mono, 16 kHz; (2) trim leading/trailing near-silence and re-pad to a fixed margin, identically for bona fide and spoof; (3) loudness-normalize to a fixed integrated loudness (EBU R128) and peak-limit; (4) optionally pass through a telephony codec, applied to both classes when used.

The front-end's configuration is hashed into a short fingerprint (640b0644ae for every result in this paper) that is stamped on every reported number; results with different fingerprints are never compared directly. After every model change, we re-run

the silence-only ablation as a permanent gate: EER on silence-only input must land within roughly five points of 50%; below 40% indicates a reintroduced shortcut and blocks trusting any other number from that run. We treat this front-end and gate, not a specific model, as the paper's core methodological contribution: it is dataset-agnostic and directly reusable by other ASVspoof 2019 LA work.

B. Signal 1: Synthetic-Voice Detection

We use AASIST [2] with publicly released pretrained weights as the acoustic spoof-detection signal, run through the front-end above on a sliding 4.04 s window with an exponential moving average (EMA) for the streaming path. We did not train this model ourselves; all numbers in Section V are for the published checkpoint evaluated through our protocol. Section VIII describes our plan to replace it with a model fine-tuned on modern generators.

C. Signal 2: Speaker Consistency

A caller enrolls with roughly 6 s of speech at session start. We embed both the enrollment audio and each subsequent window with a WavLM x-vector model (wavlm-base-plus-sv [7]) and compare by cosine similarity. Similarity is mapped to a risk score by a Platt-scaled (logistic) calibrator, fit on genuine/impostor trial pairs built from ASVspoof 2019 LA's real speaker identities and verified, on a speaker-disjoint held-out split the calibrator never trained on, to generalize to unseen speakers (Section VII) -- replacing an earlier linear interpolation between two hand-picked thresholds, which

remains a fallback when no fitted calibrator is available. Enrollment or scoring windows shorter than 2.5 s are rejected as insufficient rather than scored, because we found that short clips produce spuriously high similarity regardless of speaker identity (Section VII).

D. Signal 3: Prosody and Behavioral Biometrics

We extract 13 prosodic and voice-quality features per window using Praat/Parselmouth: F0 mean, standard deviation, range and slope; voiced fraction; jitter (local, RAP); shimmer (local, APQ5); harmonics-to-noise ratio; intensity variability; pause ratio; and short-term energy flux. A logistic regression classifier, fit on an ASVspoof 2019 LA training subset, maps these 13 features to a risk score. This signal is intentionally weak on its own (Section V); its role is to contribute evidence that is largely independent of the acoustic spoof-detection failure modes above.

E. Signal 4: Call Context

A deterministic rule set scores non-acoustic call metadata: whether the caller ID matches a known contact, the requested transaction amount (banded), the channel, and any prior flags on the session. This is the only signal not derived from audio, and it is the one most directly under an attacker's non-control - an attacker with a perfect voice clone still cannot make an unrecognized number look recognized.

F. Risk Fusion

Let S be the subset of {spoof, speaker, prosody, context} available for a given window (a signal is absent before enrollment, or if a scoring window is too short), with base weights $w(\text{spoof})=0.45$, $w(\text{speaker})=0.35$, $w(\text{context})=0.20$, $w(\text{prosody})=0.25$. The fused score is a weighted mean over present signals with weights renormalized to sum to one:

$$\text{fused} = \sum_{s \in S} [ws / \sum_{s' \in S} ws'] \cdot \text{score}_s$$

We additionally apply a lone-signal floor: if any single present signal reaches an alarm threshold $\tau(\text{alarm})=0.80$, the fused score is floored at $\tau(\text{floor})=0.68$ regardless of the weighted mean, so that one strongly suspicious signal cannot be diluted away by several weakly reassuring ones. Section V-E shows this rule firing on real data. On the streaming path, each signal is additionally smoothed with its own EMA constant, tuned per signal by observed volatility (Section VII); the file-based analysis path used for all results in this paper reports raw, unsmoothed per-clip scores.

G. Policy Engine

A fused score is mapped to an action through a per-scenario threshold table (Table I), reflecting that the acceptable false-escalation rate is not the same for a routine support call as for a large transfer or a privileged account change. The policy engine also emits plain-language reason codes built from the individual signals and context (e.g., “speaker mismatch: similarity 0.33 vs. enrolled voice (low)”, “caller number not recognised”, “large transfer requested (INR 75,000)”) and a recommendation string. Thresholds are documented operating points, not the output of a calibration procedure (Section VI-A).

TABLE I. POLICY THRESHOLDS BY CALL SCENARIO (FUSED SCORE, 0-1)

Scenario	Verify above	Escalate above
Routine support call	0.55	0.75
Transaction / transfer	0.40	0.65
Privileged change	0.30	0.55

H. Privacy-Preserving Audit Log

Every action change is written to an append-only JSONL log containing per-signal scores, the fused score, the action, the reason codes, model and front-end version, and a git commit hash for provenance. The log implementation explicitly rejects any write containing an audio, waveform, transcript, or embedding key, checked recursively, so the audit trail cannot silently grow to include raw biometric or content data. This is the system's privacy design, not merely a policy statement: it is enforced by the logging function's write path.

I. System Interfaces

A session-oriented REST and WebSocket API exposes session creation, enrollment, context updates, single-clip analysis, streaming, audit retrieval, and an acknowledge action for the human operator. A browser-based operator console renders the decision banner, a fused-risk timeline, four live signal meters, reason codes, an editable call-context panel, and the audit trail, so the same decision logic used for the results in this paper is directly demonstrable end to end.

IV. EXPERIMENTAL SETUP

A. Datasets and Splits

We use the official ASVspoof 2019 LA protocol files exclusively: the training partition for model fitting where applicable, the dev partition (attacks A01-A06, “known”) for threshold selection, and the eval partition (attacks A07-A19, “unseen”) as an in-domain test. We additionally use the In-the-Wild corpus [1] as a cross-dataset test set that is never trained on and never used to select a threshold. A home-made random split is never used, since it would leak speakers and attack types across partitions. For the pilot multilingual study (Section V-F), we generate synthetic attacks in five languages with an open multilingual TTS model [9]; no bona fide multilingual speech is included in the numbers reported here (Section VI-B).

B. Metrics

We report Equal Error Rate (EER) as the headline metric, computed at the standard operating point where false-accept and false-reject rates are equal, under the convention that higher scores mean more likely bona fide. We additionally report per-attack EER on the eval partition, scoring each attack's spoof clips against all bona fide clips separately, so a single easy attack cannot mask failure on the others. We do not report classification accuracy as a headline number, since the class prior is roughly 9:1 bona fide:spoof and accuracy is misleading at that ratio. For the multilingual pilot we report synthetic recall -- the fraction of generated attacks scored above the system's synthetic-voice threshold (0.65) -- since no bona fide

multilingual set is yet available to compute a proper EER (Section VI-B).

C. Systems Compared

We compare a Generation-1 baseline (a 61-dimensional handcrafted acoustic feature set feeding a gradient-boosted-tree ensemble of XGBoost, LightGBM and CatBoost) against AASIST with published, pretrained weights, both evaluated through the identical front-end described in Section III-A. We also implemented, but did not train to convergence, RawNet2, AASIST-L, and an SSL-AASIST variant (frozen wav2vec2/XLS-R front-end with a trainable AASIST graph back-end); a two-epoch sanity run of AASIST on our local GPU (dev EER 34.1% $\rightarrow$ 26.6% over two epochs) confirms the training loop is implemented correctly, but is not a converged model and is not used for any other result in this paper (Section VII).

D. Hardware and Reproducibility

All evaluation was run on a single consumer GPU (6 GB VRAM). Every reported row carries a provenance record (train set, test set, model, seed, front-end fingerprint, score direction, git commit, timestamp), written automatically by the evaluation harness, so that any number in this paper can be traced to the exact code and data that produced it.

V. RESULTS

A. Silence-Shortcut Ablation

Table II confirms the shortcut identified in Section III-A is closed after the front-end fix: both the Generation-1 baseline and AASIST score close to chance (50%) on silence-only input, whereas the original, unprotected pipeline scored 13-15% EER on the same kind of input.

TABLE II. SILENCE-ONLY ABLATION AFTER THE ANTI-SHORTCUT FIX (CHANCE = 50%)

System	Silence-only EER (%)
Handcrafted + GBM ensemble (Gen-1)	59.0
AASIST (published weights)	41.3

B. In-Domain and Cross-Dataset Generalization

Table III reports our headline result. AASIST, evaluated honestly through our front-end, scores 9.6% EER on known attacks and 23.3% on unseen attacks in-domain -- both markedly higher than the sub-1% figures typically reported for this model family, because our front-end removes the artifact those numbers partly relied on. More importantly, cross-dataset EER on In-the-Wild is 35.8%: substantially worse than in-domain, and far from a deployable figure, though considerably better than our own Generation-1 baseline's 58.2% (close to chance).

TABLE III. DETECTION PERFORMANCE THROUGH THE ANTI-SHORTCUT FRONT-END

System	Dev (%) known	Eval (%) unseen	ITW (%) cross-dataset
Handcrafted + GBM (Gen-1)	7.0	33.4	58.2
AASIST (published weights)	9.6	23.3	35.8

C. Per-Attack Generalization

Table IV breaks the unseen-attack (eval) result down by generator. AASIST is near-perfect on A08/A09 (waveform-concatenation and neural-vocoder attacks close to its training distribution) but fails on A10 and A12 (52.1% and 43.7% EER respectively) -- unfamiliar neural TTS/VC systems. This spread, within a single in-domain test partition, is itself evidence against treating a single aggregate EER as representative of real-world behavior.

TABLE IV. PER-ATTACK EER (%), EVAL PARTITION (UNSEEN ATTACKS), AASIST

A07	A08	A09	A10	A11	A12	A13
40.0	1.6	0.7	52.1	17.0	43.7	17.7
A14	A15	A16	A17	A18	A19	
8.0	39.0	13.7	13.7	12.4	13.4	

D. Prosody Signal, Standalone

The prosody/behavior logistic-regression classifier (Section III-D) scores 26.3% dev EER on its own -- weak by design, and weaker than the acoustic signal in isolation. We report it because its role in this paper is as a contributing, partly independent fusion input, not as a standalone detector; Section V-E shows it contributing correctly (low) on a genuine call.

E. System-Level Worked Example

Table V reports a concrete, reproducible case: a session is enrolled with a genuine speaker, a benign call segment is scored, and then a call using a different, modern neural-TTS-generated voice is scored with the call context set to an unrecognized caller and a large requested transfer (INR 75,000). The acoustic spoof signal alone (29/100) is well below the system's own synthetic-voice threshold (65/100) -- taken in isolation, the acoustic detector would not flag this call. The speaker-consistency signal, however, measures a cosine similarity of 0.33 against the enrolled voice, mapping to a risk of 99/100 and triggering the lone-signal floor (Section III-F); combined with the context signal, the fused score reaches 68/100 and the system escalates, with reason codes including the low speaker-similarity score, the unrecognized caller number, and the transfer amount. This is, to our knowledge, the clearest available illustration of the paper's central design argument: a signal the acoustic detector misses is caught by an orthogonal signal, and the fusion and floor rules are precisely what make that possible.

TABLE V. WORKED EXAMPLE: PER-SIGNAL RISK (0-100) AND DECISION

Call	Spoof	Spk.	Pros.	Ctx.	Decision (fused)
Benign	0	44	49	0	ALLOW (22)
Cloned voice*	29	99	20	65	ESCALATE (68)

* Modern neural-TTS clone, unrecognized caller, INR 75,000 transfer requested.

The benign call's speaker-consistency reading (44/100) is not zero: two different recordings of the same person are never a perfect cosine match, and the calibrated mapping (Section III, Signal 2) reflects that natural variability rather than collapsing it to zero the way an uncalibrated threshold would. It stays comfortably inside ALLOW for every scenario in Table I.

For comparison, we also tested a second, independently trained, publicly available deepfake-audio classifier as an alternative acoustic signal. It performed worse than our front-end-corrected AASIST on both ASVspoof spoof clips and, notably, missed the modern neural-TTS clone in Table V entirely -- reinforcing that the gap is not specific to AASIST, and that model-shopping for a single better acoustic detector is not, on present evidence, a substitute for fusion.

F. IndicCall-Eval: A Multilingual Pilot

SIH26104 explicitly asks for Indian-language support. Rather than assume transfer from an English-trained detector, we measured it. We generated short bank-call-style synthetic-speech attacks with an open multilingual TTS model [9] in English, Hindi, Tamil, Bengali, and Marathi (40 clips per language, native script, half passed through a randomly chosen telephony codec), and scored every clip with the same AASIST-based detector and threshold used throughout this paper. Table VI reports synthetic recall -- the fraction of generated attacks flagged -- per language.

TABLE VI. INDICCALL-EVAL PILOT: SYNTHETIC-SPEECH RECALL BY LANGUAGE (N=40/LANGUAGE, THRESHOLD 0.65)

Language	Synthetic recall (%)
Marathi	85.0
Hindi	57.5
Tamil	55.0
Bengali	37.5
English	7.5

The result is counter-intuitive and, we believe, informative: the detector, trained only on English ASVspoof 2019 data, flags Indian-language synthetic speech more often than English synthetic speech from the same TTS system. We interpret this as evidence that non-English synthesis sits further, in acoustic-feature space, from the "natural English speech" distribution the model implicitly learned as its bona fide class, so its artifacts are more likely to trip a detector tuned on that distribution -- not as evidence of genuine multilingual robustness. Section VI discusses why this is a recall measurement, not an EER, and what would be needed to turn it into one.

VI. DISCUSSION

A. Findings

The silence-shortcut finding generalizes beyond this project. Any evaluation on ASVspoof 2019 LA that does not equalize leading/trailing silence and loudness across classes is at risk of the same inflation we measured (13-15% EER from silence alone). Because this is a property of a widely used public corpus rather than of our code, we believe the anti-shortcut front-end and the permanent silence-ablation gate (Section III-A) are directly useful to other work using this corpus, independent of our system.

Fusion compensates for, but does not fix, a weak acoustic signal. The worked example in Section V-E shows fusion recovering a miss, but this depended on the speaker-consistency signal being strong evidence in that specific case (a low-similarity voice clone). An attacker who additionally spoofed or evaded the caller-ID check and used a clone similar enough to pass speaker verification would not be caught by this example's mechanism; we do not claim the current signal set is sufficient against an adaptive attacker, only that it catches more than any one signal alone on the cases we measured.

The IndicCall-Eval asymmetry is a recall number, not an EER, and should not be over-read. Because we do not yet have a consented bona fide multilingual speech set, Table VI cannot be converted into a proper per-language EER, and a high recall number under a single fixed threshold says nothing about the corresponding false-accept rate on genuine non-English speech. The pilot's value is narrower and more honest: it demonstrates that multilingual transfer is uneven and must be measured per language, and it motivates exactly the larger, consented resource that concurrent work such as IndicSynth [10] provides at a scale far beyond our pilot.

B. Limitations

We state these explicitly. (1) All detection numbers use a published, not self-trained, acoustic model; 35.8% cross-dataset EER is not a deployable figure. (2) Fusion weights and policy thresholds remain documented operating points, tuned by inspection. Of the three learned signals, only speaker consistency is now calibrated on held-out data (Section VII); spoof and prosody calibration were measured, not assumed, and found not to generalize past the data they were fit on, so both remain uncalibrated by deliberate choice rather than by omission. (3) The acoustic model's 4.04 s window is slower than a sub-2 s first-read target for a live call; no voice-activity-detection gate or lightweight first-pass model is implemented. (4) Alerts in the operator console are simulated; no production SMS/email/webhook sink is wired. (5) The IndicCall-Eval pilot uses only synthetic attacks from one open TTS family, at n=40 per language, and lacks a bona fide multilingual slice (Section V-F). (6) We have not measured performance under an adaptive attacker aware of the fusion logic. (7) We report EER throughout rather than min t-DCF/a-DCF, which is the ASVspoof community's preferred deployment-oriented metric; we treat adding it as future work (Section VIII).

VII. PRACTICAL CHALLENGES AND LESSONS LEARNED

We report the following as reproducibility-relevant findings in their own right, several of which are not specific to our codebase.

A corpus can leak the label through non-speech audio. Already discussed in Section III-A; this was the single most consequential finding of the project and the reason every other result in this paper carries a front-end fingerprint and a permanent ablation gate.

Self-supervised checkpoints can silently fail to load across library versions. We found that a WavLM speaker-verification checkpoint's positional-convolution layer, stored in the older `weight_g/weight_v` weight-normalization format, loaded without error but with randomly initialized weights under a newer version of the transformers library that switched to a parametrizations-based API -- silently degrading same-speaker cosine similarity, with no exception raised. The fix was to copy the raw weights from the checkpoint file into the new parametrization API explicitly at load time. We flag this because a silent numerical degradation of this kind is easy to miss without a same-speaker/different-speaker sanity check.

Short enrollment or scoring clips produce spuriously confident speaker matches. Below roughly 2.5 s, cosine similarity from the x-vector embedding became an unreliable indicator of speaker identity; we added an explicit minimum-duration gate rather than trusting the similarity score at any duration.

A single global smoothing constant destabilizes a fused, streaming decision. An early version applied one exponential-smoothing constant to every signal on the live streaming path; the prosody signal's higher natural volatility caused the fused decision to oscillate between ALLOW and VERIFY on stable audio. Per-signal EMA constants, chosen by each signal's observed volatility, resolved this; we note it as a general caution for any system fusing signals of different natural time-scales.

Model-shopping is not a substitute for fusion. We evaluated a second, independently published deepfake-audio detector as a candidate replacement acoustic signal. It underperformed our front-end-corrected AASIST on ASVspoof spoof clips and missed the modern-TTS clone used in Section V-E entirely. We take this as evidence against the strategy of searching for a single better off-the-shelf detector in place of building a system that does not depend on any one detector being sufficient.

Calibration inherits detection's generalization problem. We fit Platt and isotonic calibrators for all three learned signals (spoof, prosody, speaker) on held-out data and measured Expected Calibration Error (ECE) and Brier score before and after, rather than assuming a fit that looks good on its own fitting split is trustworthy. For spoof and prosody, both methods roughly halved ECE on the split they were fit on (ASVspoof dev) and made it worse on eval (unseen attacks) and, for spoof, worse again on In-the-Wild (cross-dataset) -- isotonic's dev ECE collapsed to within $1e-6$ of zero while its eval/cross-dataset ECE rose, a textbook overfitting signature repeated independently in two signals. We ship neither. The speaker-consistency signal is the exception: because same/different-speaker cosine-similarity geometry is comparatively stable (a

standard speaker-verification calibration problem) rather than shifting with an adversarial generator distribution, a Platt fit on ASVspoof's real speaker identities cut held-out (unseen speaker) ECE from 0.39 to 0.18 -- ship it, and Section III (Signal 2) reflects it. We read the contrast itself as evidence for this paper's thesis: the failure mode here is specifically about generalizing to unseen generators, not a generic "these scores are just uncalibrated" story that a fit would fix everywhere.

Full training does not fit commodity hardware at this corpus's scale. A single training epoch over ASVspoof 2019 LA took 50-57 minutes on our 6 GB laptop GPU; a real run (30-40 epochs) is several days, and our development machine could run only one such job at a time. A two-epoch run (Section IV-C) confirmed the training code is correct but is not a substitute for a full run; we moved full training to free cloud GPU compute (Section VIII) rather than report an under-trained model as a result.

Dataset-library churn broke a planned data-collection path mid-project. A major version of the datasets library dropped support for script-based dataset loaders, breaking our planned use of two public speaker-embedding/multilingual-speech resources. We adapted by generating speaker-conditioning vectors synthetically for one TTS path and deferring a real bona fide multilingual set to a smaller, directly consented team recording effort (Section VIII) rather than depending on an external pipeline that had just changed under us.

VIII. FUTURE WORK

Trained acoustic model. Replace the published AASIST weights with a version fine-tuned on modern neural-TTS output (including generators not present in ASVspoof 2019), using free cloud GPU compute; target under 15% cross-dataset EER, adopted only if it beats the current published-weights baseline under the identical evaluation protocol of Section III-A.

Calibration. Speaker consistency is now calibrated (Section VII); spoof and prosody were measured and found not to generalize with a naive dev-set fit -- worth another pass with a fitting set that spans both known and unseen attacks, or a domain-conditional calibrator, neither attempted here. The policy thresholds of Table I and the context signal remain uncalibrated operating points.

Latency. Add a voice-activity-detection gate and a lightweight first-pass model to meet a sub-2 s first-read target, alongside the current 4.04 s-window model.

Consented multilingual bona fide data. Extend IndicCall-Eval with a small, consented, recorded bona fide slice per language so that Table VI's recall numbers become proper per-language EERs; scale toward resources such as IndicSynth [10].

Deployment-oriented metrics. Report $\min$ t-DCF/a-DCF alongside EER, and add ASVspoof 2021 DF as a second, held-out cross-dataset test.

Adaptive-attacker evaluation. Measure system behavior when an attacker has knowledge of the fusion logic and specifically targets the weakest present signal.

Production hardening. Real alert delivery (SMS/email/webhook), a gRPC streaming interface and client SDK, edge/on-device inference, and a model card.

IX. CONCLUSION

We set out to build a voice-clone detector and found, first, that a widely used benchmark for this task can be partially solved by an artifact unrelated to synthesis, and second, that even after removing that artifact, a strong published detector generalizes poorly enough (35.8% cross-dataset EER) that it should not be deployed alone. Rather than continue searching for a single better classifier, we built VoiceGuard around the opposite premise: fuse several weak, partly independent signals -- acoustic, speaker-consistency, prosodic, and contextual -- into a policy-driven, auditable decision, and keep a human in the loop. We showed this design concretely recovering a case the acoustic signal alone would have missed, and we reported, alongside our results, the specific engineering failures encountered building the system, in the belief that an honest account of what did not work the first time is as reusable to the field as the numbers that did.

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